

Session Transcript: Catmull-Rom Grouping and Derivative

User + Codex

2026-02-28

Conversation (Condensed)

User: Here are equations:

$$\begin{aligned}A1 &= \frac{t_1 - x}{t_1 - t_0} p_0 + \frac{x - t_0}{t_1 - t_0} p_1 \\A2 &= \frac{t_2 - x}{t_2 - t_1} p_1 + \frac{x - t_1}{t_2 - t_1} p_2 \\A3 &= \frac{t_3 - x}{t_3 - t_2} p_2 + \frac{x - t_2}{t_3 - t_2} p_3 \\B1 &= \frac{t_2 - x}{t_2 - t_0} A1 + \frac{x - t_0}{t_2 - t_0} A2 \\B2 &= \frac{t_3 - x}{t_3 - t_1} A2 + \frac{x - t_1}{t_3 - t_1} A3 \\C &= \frac{t_2 - x}{t_2 - t_1} B1 + \frac{x - t_1}{t_2 - t_1} B2\end{aligned}$$

User: x is a variable.

User: Regroup C so that it has 4 additions grouped by p_1, p_2, p_3, p_4 .

Assistant (result):

$$C = k_1 p_1 + k_2 p_2 + k_3 p_3 + k_4 p_4.$$

User: Create a small standalone C program to verify this.

Assistant: Added `verify_catmull_grouping.c` and validated grouped form numerically against the original nested form.

User: Before grouping, group each coefficient by x for analytical derivative.

Assistant: Provided polynomial-in- x coefficients and derivative form:

$$\frac{dC}{dx} = \sum_{i=1}^4 k'_i(x) p_i.$$

User: Just multiply differences.

Assistant: Rewrote coefficients using multiplied-difference factors only.

User: Yes, add.

Assistant: Extended verifier to include analytical derivative checks and finite-difference validation.

Multiplied-Difference Grouping Used

$$C(x) = k_1(x)p_1 + k_2(x)p_2 + k_3(x)p_3 + k_4(x)p_4$$

$$\begin{aligned} k_1 &= \frac{(t_2 - x)(t_2 - x)(t_1 - x)}{(t_2 - t_1)(t_2 - t_0)(t_1 - t_0)} \\ k_2 &= \frac{(t_2 - x)(t_2 - x)(x - t_0)}{(t_2 - t_1)(t_2 - t_0)(t_1 - t_0)} + \frac{(t_2 - x)(t_2 - x)(x - t_0)}{(t_2 - t_1)(t_2 - t_1)(t_2 - t_0)} + \frac{(x - t_1)(t_3 - x)(t_2 - x)}{(t_2 - t_1)(t_2 - t_1)(t_3 - t_1)} \\ k_3 &= \frac{(t_2 - x)(x - t_0)(x - t_1)}{(t_2 - t_1)(t_2 - t_1)(t_2 - t_0)} + \frac{(x - t_1)(x - t_1)(t_3 - x)}{(t_2 - t_1)(t_2 - t_1)(t_3 - t_1)} + \frac{(x - t_1)(x - t_1)(t_3 - x)}{(t_2 - t_1)(t_3 - t_1)(t_3 - t_2)} \\ k_4 &= \frac{(x - t_1)(x - t_1)(x - t_2)}{(t_2 - t_1)(t_3 - t_1)(t_3 - t_2)} \end{aligned}$$

K_i Grouped by Powers of x

Define:

$$\Delta_{10} = t_1 - t_0, \quad \Delta_{21} = t_2 - t_1, \quad \Delta_{32} = t_3 - t_2, \quad \Delta_{20} = t_2 - t_0, \quad \Delta_{31} = t_3 - t_1.$$

For each $i \in \{1, 2, 3, 4\}$:

$$k_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i.$$

$$\begin{aligned} a_1 &= -\frac{1}{\Delta_{21}\Delta_{20}\Delta_{10}}, & b_1 &= \frac{2t_2 + t_1}{\Delta_{21}\Delta_{20}\Delta_{10}}, \\ c_1 &= -\frac{t_2^2 + 2t_1 t_2}{\Delta_{21}\Delta_{20}\Delta_{10}}, & d_1 &= \frac{t_1 t_2^2}{\Delta_{21}\Delta_{20}\Delta_{10}}. \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{\frac{1}{\Delta_{10}} + \frac{1}{\Delta_{31}}}{\Delta_{21}^2}, \\ b_2 &= \frac{-\frac{t_0 + 2t_2}{\Delta_{10}} - \frac{t_1 + t_2 + t_3}{\Delta_{31}}}{\Delta_{21}^2}, \\ c_2 &= \frac{\frac{2t_0 t_2 + t_2^2}{\Delta_{10}} + \frac{t_1 t_2 + t_1 t_3 + t_2 t_3}{\Delta_{31}}}{\Delta_{21}^2}, \\ d_2 &= \frac{-\frac{t_0 t_2^2}{\Delta_{10}} - \frac{t_1 t_2 t_3}{\Delta_{31}}}{\Delta_{21}^2}. \end{aligned}$$

$$\begin{aligned}
a_3 &= \frac{-\frac{1}{\Delta_{20}} - \frac{1}{\Delta_{32}}}{\Delta_{21}^2}, \\
b_3 &= \frac{\frac{t_0+t_1+t_2}{\Delta_{20}} + \frac{t_3+2t_1}{\Delta_{32}}}{\Delta_{21}^2}, \\
c_3 &= \frac{-\frac{t_0t_1+t_0t_2+t_1t_2}{\Delta_{20}} - \frac{2t_1t_3+t_1^2}{\Delta_{32}}}{\Delta_{21}^2}, \\
d_3 &= \frac{\frac{t_0t_1t_2}{\Delta_{20}} + \frac{t_1^2t_3}{\Delta_{32}}}{\Delta_{21}^2}.
\end{aligned}$$

$$\begin{aligned}
a_4 &= \frac{1}{\Delta_{21}\Delta_{31}\Delta_{32}}, & b_4 &= -\frac{2t_1+t_2}{\Delta_{21}\Delta_{31}\Delta_{32}}, \\
c_4 &= \frac{t_1^2+2t_1t_2}{\Delta_{21}\Delta_{31}\Delta_{32}}, & d_4 &= -\frac{t_1^2t_2}{\Delta_{21}\Delta_{31}\Delta_{32}}.
\end{aligned}$$

Hence:

$$k'_i(x) = 3a_ix^2 + 2b_ix + c_i, \quad \frac{dC}{dx} = \sum_{i=1}^4 k'_i(x) p_i.$$

Derivative structure:

$$\frac{dC}{dx} = k'_1(x)p_1 + k'_2(x)p_2 + k'_3(x)p_3 + k'_4(x)p_4,$$

with each $k'_i(x)$ obtained via product rule on its 3 linear factors.

Derivative Equations

$$\begin{aligned}
k'_1(x) &= \frac{(-1)(t_2-x)(t_1-x) + (t_2-x)(-1)(t_1-x) + (t_2-x)(t_2-x)(-1)}{(t_2-t_1)(t_2-t_0)(t_1-t_0)} \\
k'_2(x) &= \frac{(-1)(t_2-x)(x-t_0) + (t_2-x)(-1)(x-t_0) + (t_2-x)(t_2-x)(1)}{(t_2-t_1)(t_2-t_0)(t_1-t_0)} \\
&\quad + \frac{(-1)(t_2-x)(x-t_0) + (t_2-x)(-1)(x-t_0) + (t_2-x)(t_2-x)(1)}{(t_2-t_1)(t_2-t_1)(t_2-t_0)} \\
&\quad + \frac{(1)(t_3-x)(t_2-x) + (x-t_1)(-1)(t_2-x) + (x-t_1)(t_3-x)(-1)}{(t_2-t_1)(t_2-t_1)(t_3-t_1)} \\
k'_3(x) &= \frac{(-1)(x-t_0)(x-t_1) + (t_2-x)(1)(x-t_1) + (t_2-x)(x-t_0)(1)}{(t_2-t_1)(t_2-t_1)(t_2-t_0)} \\
&\quad + \frac{(1)(x-t_1)(t_3-x) + (x-t_1)(1)(t_3-x) + (x-t_1)(x-t_1)(-1)}{(t_2-t_1)(t_2-t_1)(t_3-t_1)} \\
&\quad + \frac{(1)(x-t_1)(t_3-x) + (x-t_1)(1)(t_3-x) + (x-t_1)(x-t_1)(-1)}{(t_2-t_1)(t_3-t_1)(t_3-t_2)} \\
k'_4(x) &= \frac{(1)(x-t_1)(x-t_2) + (x-t_1)(1)(x-t_2) + (x-t_1)(x-t_1)(1)}{(t_2-t_1)(t_3-t_1)(t_3-t_2)}
\end{aligned}$$

Verification Outcome

From `verify_catmull_grouping.c` run:

- PASS: 200000 random tests
- max abs error: 3.411×10^{-13}
- max abs derivative error: 3.233×10^{-7}

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